AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2014

General Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL Questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 5 Marks)

a.

i. Simplify $\frac{(x+y)^3}{x^2-y^2} \div \frac{(x+y)^2}{x^4-y^4}$ (3 Marks)

ii. If $x = \sqrt{2} - 1$, then express $\frac{1}{x}$ in the simplest form. (2 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

b. If $(a+b)^2 = 49$ and $a^2 + b^2 = 25$, then find $a - b$. (5 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 5 Marks)

a. Completely factorize $x^3 + 3x^2 - 4x - 12$ given that $x - 2$ is one of its factors. (5 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

b.

i. Factorize completely $27x^3 - 1$ (2 Marks)

ii. Verify that $x = 0$ and $x = \frac{-b}{a}$ are zeros of polynomial $P(x) = ax^2 + bx$ (3 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 4 Marks)

a. Simplify $\left[\frac{x-y}{x^3y-xy^3}\div\frac{1}{x+y}\right]\left(x^2y^2\right)$ (4 Marks)

b.

i. Find the square root of x^4-8x^2+16 (3 Marks)

ii. Find the highest common factor (HCF) of $(x-4)\left(x^2+4x+16\right)$ and $(x-4)^2$ (1 Mark)

Q.4. (Total 5 Marks)

i. Solve $3x + \frac{2}{3} = \frac{8}{3}$ (2 Marks)

ii. Find the solution set of $3 - 2x < 5, x \in R$. Also represent the solution set on the number line. (3 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 5 Marks)

a. Find the values of x for $6x^2 - 31x = -35$

b. Solve $x^2 - x = 12$ by completing square method.

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a. Find X if $3X + \frac{1}{2}\begin{bmatrix} 6 & 8 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix}$ (5 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

b. If $A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$, then find the following:

i. additive inverse of A (1 Mark)

ii. transpose of A (1 Mark)

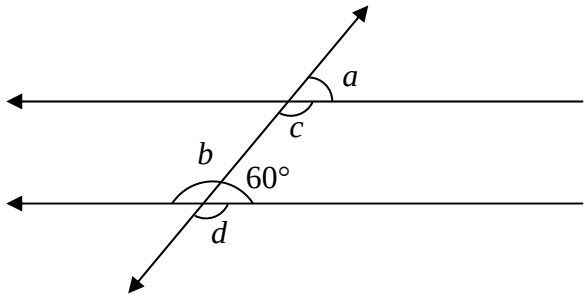
iii. determinant of A (1 Mark)

iv. additive identity of A (1 Mark)

v. multiplicative identity of A (1 Mark)

Q.7. (Total 5 Marks)

In the given figure find *a*, *b*, *c* and *d*. What is the sum of the interior angles on the same side of the transversal?

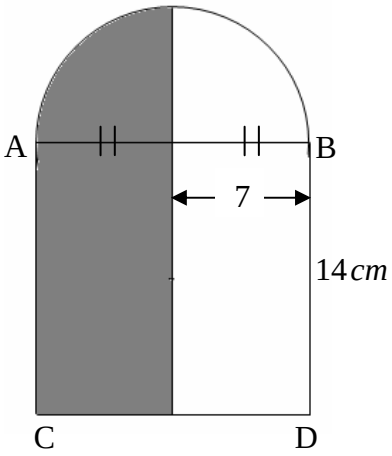


Q.8. (Total 4 Marks)

Draw an equilateral triangle and prove that its angle bisectors are concurrent.

Q.9. (Total 4 Marks)

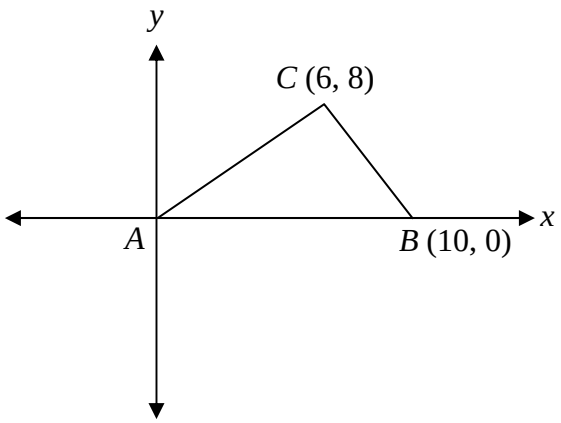
The diameter $\overline{AB}$ of a circle is the side of a square $ABCD$ as shown in the figure below. Find the area of the shaded portion.



Note: $\pi = \frac{22}{7}$

Q.10. (Total 3 Marks)

Prove that in the given diagram, ΔABC is an isosceles triangle.



NOT TO SCALE

Please use this page for rough work

Please use this page for rough work